Augmenting a Maquette of Bergen-Belsen Concentration Camp with Prisoner Artwork

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Introduction

Most Holocaust and Nazi persecution memorial sites use physical models, i.e. maquettes, to explain to visitors the spatial organisation of the historical site (Wraight 2023). As such these maquettes represent a more general trend of capturing reality through models to communicate some of its salient features, i.e. physical space. They serve as a "substitute for the objectively intangible world," in this case structures of the past that no longer exist, and therefore can become "epistemological objects" for creating or conveying (historical) knowledge (Meister / Ma 2024: 8). Visitors of Nazi persecution memorial sites often start by seeking an overview of the site's spatial and historical situation. Currently, many memorial sites offer physical models for an understanding of the historical layout,[1] while exhibitions present detailed information i.e. on the camp's structure, development and the victims' experience. However, these elements are separate and require the visitors' ability to not only process the new information but also combine it mentally.

Today, the Bergen-Belsen memorial is landscaped as a park-like area of the former campgrounds.[2] As such it does not resemble the topography of the camp, making it difficult to connect to the site's history. We have earlier made the historical site accessible through mobile applications, allowing users to explore the memorial landscape's history through Augmented and Virtual Reality (XR) following an active inquiry-based learning paradigm called Future Memory (Verschure / Wierenga 2021). Here we discuss a further extension of the Future Memory approach to enhance a physical maquette of the Bergen-Belsen concentration camp, constructed in 1990 (Figure 1), with witness testimonies and artworks as part of the MEMORISE exhibition "Showing the Unspeakable: Testimonies from Bergen Belsen" (November, 2024).

Figure 1. The marquette at Bergen-Belsen Memorial's educational centre.

In Bergen-Belsen, a physical maquette is used to visualise the structures and spatial dimensions of the former camp. The maquette, located in the educational centre, depicts the concentration camp structure of September 1944 based on an aerial photograph.[3] It is a starting point for guided tours and many independent visitors to get an overview of what the camp used to look like. While guides can serve as narrators and interpreters of historical information and thus make connections between the camp model and camp history, this contextual information is not immediately available for independent visitors.

Displaying only physical structures and landscapes, devoid of visualising the overcrowded conditions, the model could even convey a distorted impression of the reality of the camp among visitors, which could lead to "historical falsehood" (Lopez-Menchero / Grande 2011 : 4 - 5.). Furthermore, while documents, artworks and testimonies are displayed in the main exhibition to inform about the chronological and functional development of the camp, these materials are not contextualised in a spatial overview of the former camp. Similar to other memorial sites, the outside area of Bergen-Belsen is not entirely accessible. Therefore, a digital enhancement of the camp maquette combines geographical, structural, and historical presentation of the site to all visitors supporting their agency in discovering historical narratives.

Augmenting the Maquette of Bergen-Belsen concentration camp

The AR application presented in this paper, called Future Memory Maquette Explorer, FMme, uses object recognition technology to superimpose effects and localised points of interest (PoI) onto the physical model. In the context of the 2024 MEMORISE exhibition "Showing the Unspeakable" the application allows users to explore prisoner artworks. The user can tap on the PoIs to see the artworks overlaid on a 1944 3D model of the camp, and read information about the locations, artworks and artists.[4] MEMORISE is a Horizon Europe project aiming to develop new digital storytelling environments that are capable of engaging users to interact with the heritage of Nazi persecution. MEMORISE focuses on the personal experiences of the victims, recorded in diaries, testimonies and artworks, thus communicating their memories in intuitive and modern ways.

The exhibition used prisoner artworks to allow visitors to explore subjective experiences of former prisoners through their visual testimonies.[5] For a long time, drawings by prisoners were not recognized as historical sources. Today we see them as subjective testimonies of camp reality. They can help to make details of life and suffering visible, and the use of different techniques and styles allows a personal approach to the history of Nazi persecution which complements other sources of information. Finally, such drawings, especially when enriched with information about their relation to physical reality, their creators and their biographies, can help convey the human dimension of history and thus provide a productive contrast to the "objective" materialisation of historical information in a camp model.[6]

The FMme tool offers an opportunity for visitors to independently explore visual and written testimonies of former prisoners with AR while getting an overview of the former concentration camp's structure, connecting space with individual experiences and narratives. This presentation will discuss the state-of-the-art of AR technology at memorial sites, the development of the AR application and initial feedback from visitors of the memorial.

State of the Art: AR Applications at Memorial Sites of Nazi Persecution

AR applications have been increasingly adapted to memorial sites of Nazi persecution.[7] The Bergen-Belsen memorial collaborated with Future Memory Foundation and Eodyne Systems to develop The Bergen-Belsen AR landscape guide, mainly used in educational settings for the past 12 years (Future Memory Foundation 2024).[8] Furthermore, the Dachau Concentration Camp Memorial Site has collaborated with ZAUBAR on two different applications that use AR at the memorial

grounds (ZAUBAR n.d.).[9] Moreover, SPUR.lab has developed an AR application HORIZON that connects georeferenced victims' histories of Brandenburg to the user's geographical location, visualising an explorable map of connections (SPUR.lab n.d.).

Figure 2. An overlaid drawing by Louis Asscher in the AR application (access rights NIOD).

While most of the AR applications in use at Nazi persecution memorial sites focus on augmenting the physical location of the user at memorial grounds, this application augments the model of the historical site. So far, the only site with a similar approach is the Wolfenbüttel prison memorial.[10] Their AR application overlays headings on the buildings, and users can read more information, explore annotated documents, and view scanned interiors using handheld VR. As access to the still functioning prison is restricted, the application makes the buildings digitally accessible for the memorial's visitors. Wolfenbüttel's AR application differs in the contextual information it provides and user experience. In FMme the maquette serves as a frame for the victims' experiences and information, that visitors can explore. The artworks are augmented in the model and this way geographical context is offered to the depicted situations adding an additional layer of context and memory scaffolding (Verschure / Wierenga 2021). The focus of the current prototype is the prisoner artists' perspective on camp life and users are invited to follow this gaze (Figure 2).

Methodology

We developed this app using the Unity game engine in combination with the Vuforia platform for Augmented Reality software. We configured the app to recognise the maquette as a 3D object, on top of which the different digital elements are placed, accurately geolocalised. This included georeferenced artworks of prisoners of the concentration camp, biographies of the artists as well as topographical information of the camp's structure. In its current version, the app is distributed privately for installation only on the memorial site's devices. This includes three iPad Pros placed on fixed but articulated stands around the maquette (Figure 3). In the future, we plan on making the app available for download on personal devices of the visitors.

Figure 3. On the iPad screen, the application recognises the maquette and superimposes location points of the artwork and augmented fog over the maquette.

During the course of the exhibition we will gather initial qualitative user feedback on the users' experience of the developed application. We can later use this information to improve the selection of content available for users, as well as providing personalised recommendations. These initial results will be reported in the presentation.

Conclusion

We will present the results of user activity and feedback during the presentation in case of acceptance. Orientation and overview are important aspects for memorial site visitors, in particular where physical camp structures no longer exist. In many sites, models provide geographical orientation while exhibitions give an overview of the camp's development. Although the Future Memory XR landscape guide in Bergen-Belsen offers orientation and information for visitors on site, Future Memory Maquette Explorer can be used by all visitors regardless of the degree of the site's

accessibility. This format is available for independent use throughout the memorial's opening hours. Currently, the content selection offers context to the physical overview via the AR application and it can be thematically changed with little effort to emphasise victim groups, source formats (pictures, texts, maps, etc.), events or other topics through its database with curated historical information and sources. Finally, the evaluation of the current version might show how visitors engage with subjective stories and expressions of prisoners and survivors. While concern towards using deep technology at memorial sites has been rightly expressed, this developed technology aims to address the need for accessibility and individualised learning building on a scientific grounded approach.

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Notes

1. Some of these models were purposely constructed by survivors to bear witness to the crimes committed at these sites.
2. The area is dotted with some remains of foundations, and maquettes and monuments erected post Second World War
3. The model was constructed in April 1990 in connection with the then expanded Document Building and a new exhibition. The aerial reconnaissance photograph was taken by the Royal Air Force in September, 1944.
4. FMme provides a framework that can be further adapted and curated to exhibit different source materials and themes related to the historical site.
5. There are very few photographs that depict the concentration camp during its operation: most photographs were taken by the British troops after liberating the camp in April 1945.
6. This follows the Future Memory paradigm of conveying the chronotype of historical narratives through combining space, objects, personal testimonies, facts and context.
7. See for instance: Wagner 2019; Verschure & Wierenga 2021; Benardou & Droumpouki 2023; Challenor & Ma 2023
8. In this application visitors to the memorial grounds can view a geolocalised 3D model of the former camp in different time layers, spatially annotated with written and visual sources like diary entries, artwork and photographs. The application is expanded to a number of museums and memorial sites across Europe.
9. The Liberation AR application superimposes liberation photos on location at the memorial grounds, while the ARt Dachau AR application overlays prisoner and descendant artwork. See more at: https://www.kz-gedenkstaette-dachau.de/en/history-online/art-concentration-camp-in-drawings/e and https://www.kz-gedenkstaette-dachau.de/en/history-online/the-liberation/
10. At Wolfenbüttel, users can use an iPad to augment a physical model of the historical structure of prison buildings using AR. Wolfenbüttel is also part of the Stiftung niedersächsische Gedenkstätten, which manages the Bergen-Belsen Memorial. See more at: https://www.youtube.com/watch?v=ObyFn8Y2RYY

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